

Sewon Park

503, Department of Statistics, Sookmyung Women's University
100, Cheongpa-ro 47-gil, Yongsan-gu, Seoul, Republic of Korea, 04310
swpark0413@gmail.com or swpark0413@sookmyung.ac.kr

RESEARCH INTEREST

My research interests lie at the intersection of Bayesian statistics and deep learning, with a particular focus on modeling tabular and time-series data. I have expertise in nonparametric Bayesian methods, synthetic data generation, time-series anomaly detection, and missing data analysis. My current research focuses on developing methods for nonparametric regression and dimension reduction.

PROFESSIONAL EXPERIENCE

Assistant Professor **September 2025 - Present**
Department of Statistics, Sookmyung Women's University, South Korea

Senior Engineer **September 2021 - August 2025**
Security Algorithm Lab, Samsung SDS, South Korea

EDUCATION

Ph.D., Statistics **March 2015 - August 2021**
Department of Statistics, Seoul National University, Seoul, South Korea
• **Advisor:** Professor Jaeyong Lee.
• **Thesis:** *Bayesian Nonparametric Regression via Overcomplete Systems with B-spline Bases*

Bachelor of Arts, Economics, Applied Statistics (Double Major) **February 2015**
Department of Economics, Yonsei University, Gangwon, South Korea

TEACHING EXPERIENCE

Instructor **Fall 2025 - Present**
Sookmyung Women's University, South Korea

- Advanced Bayesian Statistics (**Graduate**) **Fall 2026**
- Applied Data Analysis with R-INLA **Fall 2026**
- Bayesian Statistics **Spring 2026**
- Big Data Statistical Analysis **Spring 2026**
- Categorical Data Analysis **Fall 2025**
- Advanced Applied Statistics (**Graduate**) **Fall 2025**

Teaching Assistant **Spring 2015 - Spring 2019**
Seoul National University, South Korea

- Mathematical Statistics 1 **Spring 2016, Spring 2019**
- Bayesian Statistics and Lab **Fall 2018**
- Regression Analysis and Lab **Spring 2018**
- Statistical computing and Lab **Fall 2016, Fall 2017**
- Statistical Consulting and Practices **Spring 2017**
- Statistics Concept and Lab **Fall 2015**
- Introduction to Statistics **Spring 2015**

HONORS & AWARDS	Winner of the 2023 T-SM Best Paper Award	January 2024
	First prize winner in Graduate Student Paper Award (Korean Statistical Society)	
	Winter 2020	
	Brain Korea 21 Plus Scholarship (National Research Foundation)	Spring 2016 -
	Spring 2019	
	Merit-based Scholarship (Seoul National University)	Fall 2015
	Merit-based Scholarship (Yonsei University)	Fall 2012, Spring 2007

PROJECTS	Insider threat detection, Samsung SDS	January 2024 - August 2025
	Malware detection, Samsung SDS	January 2023 - December 2023
	Time-series anomaly detection, Samsung SDS	July 2022- June 2023
	Synthetic data generation, Samsung SDS	September 2021 - June 2022
	Statistical research for semiconductor manufacturing data, Samsung Electronics, DS division	Fall 2018 - August 2021
	Development of the cost estimation methods for radar systems, Agency for Defence Development (ADD)	December 2016 - November 2017

PUBLICATIONS * Corresponding author, [†] Co-first author.

Kwangmin Lee, Yeonhee Park, **Sewon Park**^{*}. (2026, July). Reflective Hamiltonian Monte Carlo: Mixing Analysis and Application to Sampling on Stiefel Manifold. In Proceedings of the 43rd International Conference on Machine Learning (ICML 2026).

Haksoo Lim, Jaehoon Lee, **Sewon Park**, Minjung Kim, and Noseong Park. (2025, December). TSGM: Regular and Irregular Time-series Generation using Score-based Generative Models. In 2025 IEEE International Conference on Big Data (BigData).

Jinsung Jeon, Jaehyeon Park, **Sewon Park**, Minjung Kim, Jeongwhan Choi and Noseong Park. (2025, April). Possibility for Proactive Anomaly Detection. In Proceedings of the ICLR Workshop on ICBINB.

Haksoo Lim, **Sewon Park**[†], Minjung Kim, Jaehoon Lee, Seonkyu Lim, and Noseong Park. (2023, October). MadSGM: Multivariate Anomaly Detection with Score-based Generative Models. In Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (pp. 1411-1420).

Sewon Park^{*}, Hee-Seok Oh, and Jaeyong Lee. (2023, October) Lévy Adaptive B-Spline Regression via Overcomplete Systems. Statistica Sinica, 33 (4).

Sewon Park^{*}, Kyeongwon Lee, Da-Eun Jeong, Heung-Kook Ko and Jaeyong Lee. (2023, May) Bayesian Nonparametric Classification for Incomplete Data With a High Missing Rate: an Application to Semiconductor Manufacturing Data. IEEE Transactions on Semiconductor Manufacturing, 36 (2), pp. 170-179.

Jayoung Kim, Chaejeong Lee, Yehjin Shin, **Sewon Park**, Minjung Kim, Noseong Park, and Jihoon Cho. (2022, August). SOS: Score-based oversampling for tabular data. In Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22), pp. 762-772.

Suji Lee, Sukjin Han, **Sewon Park**^{*}, Kyeongwon Lee, and Jaeyong Lee. (2019) Korean speech recognition using deep learning (in Korean). The Korean Journal of Applied Statistics, 32 (2), pp. 213-227.

**PAPERS IN
PREPARATION**

*Seongmin Kim, Kwangmin Lee, **Sewon Park** and Jaeyong Lee. Eigenstructure inference for high-dimensional covariance with generalized shrinkage inverse-Wishart prior. under major revision in Bayesian Analysis.*

*Kwangmin Lee, **Sewon Park**^{*}, Seongmin Kim and Jaeyong Lee. Bayesian Analysis of Spiked Covariance Models: Correcting Eigenvalue Bias and Determining the Number of Spikes. Submitted.*

*Jeunghun Oh, **Sewon Park** and Jaeyong Lee. Posterior Contraction of Lévy Adaptive B-spline Regression in Besov Spaces. Submitted.*

**TECHNICAL
SKILLS**

R (Excellent), Python (Excellent), PyTorch (Excellent), C/C++ (Advanced)